



Scientific Project

Height Measurement System Using Proximity 15 Click Sensor

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in partial fulfillment of the requirements for the degree of
Bachelor
in
**Electrical Engineering and Information
Technology**
at
Hochschule Ravensburg-Weingarten

April 30, 2025

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1 Acknowledgement

We would like to express our sincere gratitude to **Prof. Lothar Berger**, who guided us throughout this project and provided invaluable feedback and support. His expertise and encouragement were instrumental in shaping my understanding of hardware systems and their practical applications.

Additionally, we would like to thank **Hochschule Ravensburg - Weingarten**, which provided us with the necessary resources and facilities to complete this project. The support, laboratory infrastructure, and technical environment offered by the university were crucial in facilitating our research and development efforts.

We would also like to extend our heartfelt thanks to our families and friends for their unwavering support and patience throughout this journey. Their encouragement has been a constant source of motivation and inspiration during the course of this work.

2 Abstract

This scientific report presents the design and development of a Height Measurement System using the Proximity 15 Click sensor module, which incorporates the VL53L1X Time-of-Flight (ToF) sensor. The project focuses on accurately measuring object distance using laser-based ranging technology integrated with the EFM8BB52 and ESP32 microcontroller.

The report details the theoretical background of ToF measurement principles, the operation of the VL53L1X sensor, and I2C communication fundamentals. It further elaborates on the hardware setup, circuit design, firmware development, and system architecture implemented to achieve accurate and reliable measurements.

Extensive experimental validation was performed using UART-based communication testing and a measuring tape. The results demonstrate the effectiveness of the system, highlighting its ability to perform precise distance measurements under different environmental conditions.

The developed system serves as a foundational prototype for future enhancements in fields such as automation, robotics, and smart sensing applications.

3 Introduction

3.1 Objective

The primary objective of this project is to design, develop, and implement a Height Measurement System using the Proximity 15 Click sensor board based on the VL53L1X Time-of-Flight (ToF) sensor. The system aims to achieve accurate, real-time distance measurements between the sensor and target objects. This data can then be used for various industrial, consumer, and research applications requiring precise distance sensing, such as object detection, robotics, and smart automation.

The specific goals of this project are:

- To understand and utilize the VL53L1X ToF sensor's capabilities for distance measurement.
- To interface the sensor module with the EFM8BB52 microcontroller via the I2C communication protocol.
- To develop efficient firmware to collect, process, and transmit the measurement data using UART.
- To validate the system's performance through functional testing.

While significant progress was made in hardware integration and communication setup, full successful measurement of distance with EFM8BB52 microcontroller could not be achieved. The sensor remained in the firmware setup phase despite multiple troubleshooting efforts.

3.2 Challenges

During the development of this project, several technical challenges were encountered and addressed:

- **Sensor Initialization:** Properly configuring the VL53L1X sensor and ensuring correct boot sequencing proved critical. Incorrect initialization repeatedly caused the sensor to remain unresponsive, preventing complete distance measurements from being obtained.
- **I2C-UART conflicts:** Managing simultaneous I2C and UART communications on the EFM8BB52 microcontroller proved to be challenging due to hardware resource sharing and interrupt conflicts, requiring careful configuration and debugging.
- **Complex configuration:** Setting up the EFM8BB52 at the register level using Simplicity Studio demanded a thorough understanding of the microcontroller's architecture and peripherals. Achieving proper initialization sequences for both I2C and UART interfaces was particularly intricate.
- **Verification:** To ensure the correctness of the sensor readings and system behavior, an alternative verification method was employed. The VL53L1X sensor was tested separately using an ESP32 microcontroller programmed via the Arduino IDE and VL53L1X library, providing a known-good reference for system validation.

4 Literature Review

4.1 Time-of-Flight (ToF) Theory

Time-of-Flight (ToF) distance measurement is a method based on calculating the time it takes for a light signal (typically infrared) to travel from a transmitter to a target object and back to a receiver. [8]

By knowing the speed of light in air (299,792,458 meters per second), and measuring the round-trip time precisely, the distance between the sensor and the object can be accurately determined using the formula:

$$d = \frac{c \times t}{2}$$

Where:

- d = distance to the object
- c = speed of light
- t = measured round-trip time

ToF systems are widely used in applications requiring high accuracy at short-to-medium distances, including industrial sensing, smartphone cameras, robotics, and automotive safety systems.

The key advantages of ToF technology include its relative immunity to object color or reflectivity and its ability to function in various ambient lighting conditions.

4.2 VL53L1X Sensor Theory

The VL53L1X from STMicroelectronics is a second-generation ToF sensor that measures distances by emitting infrared laser pulses (Class 1 eye-safe laser) and detecting their reflections.[6]

Key features of the VL53L1X include:

- **Ranging distances up to 4 meters** in optimal conditions.
- **Single photon avalanche diode (SPAD) array** used for ultra-precise detection.
- **Integrated physical infrared filters** to suppress ambient light noise.
- **Multi-zone detection and configurable region-of-interest (ROI)** for more flexible sensing applications.

Unlike basic proximity sensors, which infer distance based on signal strength, the VL53L1X measures the actual time taken by photons to return, offering highly accurate absolute distance measurements independent of surface reflectivity or color.

The sensor communicates over the I2C bus and requires specific initialization sequences to load its firmware properly after power-up[5]. Once initialized, it can operate in either continuous ranging mode or single-shot mode depending on application needs.

4.3 I2C Communication

The Inter-Integrated Circuit (I2C) protocol is a widely used synchronous, multi-master, multi-slave, packet-switched, single-ended, serial communication bus. Developed by Philips Semiconductor (now NXP) in the 1980s, it is commonly used for connecting microcontrollers to peripheral devices like sensors, memory chips, and displays.[3][7][9]

Key characteristics of I2C include:

- **Two-wire communication:** Serial Data Line (SDA) and Serial Clock Line (SCL).
- **Addressed communication:** Each slave device has a unique 7- or 10-bit address.
- **Master-slave relationship:** Master initiates and controls all communications.
- **Open-drain lines:** Requires pull-up resistors on both SDA and SCL lines.
- **Standard operating speeds:** 100 kbit/s (Standard Mode), 400 kbit/s (Fast Mode), and higher.

In this project, the EFM8BB52 microcontroller acts as the I2C master, while the VL53L1X sensor acts as the slave. The master sends start conditions, slave addresses, register addresses, and read/write commands, ensuring synchronized data transfers for sensor configuration and distance data retrieval.

4.4 UART Communication

The Universal Asynchronous Receiver/Transmitter (UART) is a hardware communication protocol used for asynchronous serial communication between devices. Unlike synchronous protocols such as I2C and SPI, UART does not require a shared clock signal. Instead, both transmitting and receiving devices must agree on common transmission parameters such as baud rate, data bits, stop bits, and parity.[4]

Key features of UART communication include:

- **Asynchronous transmission:** Data is sent one bit at a time without a clock signal, using start and stop bits to frame each byte.
- **Full-duplex communication:** UART can transmit and receive data simultaneously through two separate lines:
 - TX (Transmit)
 - RX (Receive)
- **Configurable settings:** Baud rate (e.g., 9600, 115200 bps), parity (even/odd/none), number of stop bits, and word length are user-definable.
- **Error detection:** Basic error detection mechanisms such as parity checks and framing error detection.

In this project, the EFM8BB52 microcontroller uses its internal UART0 peripheral to transmit distance measurement data to a PC terminal for monitoring. The UART was configured with a specific baud rate to match the PC terminal settings of 115200 bps, ensuring reliable real-time data visualization.[1]

UART communication plays a crucial role during system development, debugging, and performance evaluation by allowing a simple, effective way to monitor sensor outputs and detect system behavior without intrusive testing setups.

5 Materials and Methods

5.1 Hardware Components

5.1.1 EFM8BB52 Microcontroller

The EFM8BB52 microcontroller from Silicon Labs was selected as the primary processing unit for this project. It is an 8-bit MCU based on the 8051 core architecture, offering high-speed operation, a low-power design, and integrated peripherals.[1]



Figure 1: BB52 Explorer Kit

Key features utilized in this project include:

- **I2C Master Interface (SMBus):** For communication with the Proximity 15 Click sensor.
- **UART0 Peripheral:** For sending sensor data to a serial terminal for testing and monitoring.
- **GPIO Control:** For toggling status LEDs and sensor reset pins.

The EFM8BB52 was programmed using Simplicity Studio 5 IDE, utilizing the Silicon Labs Hardware Configurator and C language-based firmware development.

5.1.2 Proximity 15 Click

The Proximity 15 Click is an add-on board from MikroElektronika, featuring the VL53L1X Time-of-Flight distance sensor by STMicroelectronics. This sensor module integrates a SPAD array (Single Photon Avalanche Diode), optics, and a laser emitter to accurately measure distances up to 4 meters.



Figure 2: Proximity 15 Click

Key features relevant to the project include:

- **High-accuracy ranging** independent of target reflectance or color.
- **I2C interface** for configuration and data retrieval.
- **Wide operating range** (ranging beyond 4 meters under ideal conditions).
- **On-board XSHUT pin** for hardware-based sensor reset control.

The module was interfaced directly with the EFM8BB52's I2C lines and powered by the 3.3V supply available on the development board.[2]

5.2 Circuit Connections

A direct wiring setup was implemented to connect the EFM8BB52 development board and Proximity 15 Click sensor.

EFM8BB52 Pin	VL53L1X Pin	Function
P1.5 (SDA)	SDA	I2C Data
P1.6 (SCL)	SCL	I2C Clock
P1.3 (XSHUT)	XSHUT	Sensor Reset
P1.4 (LED0)	-	LED Indicator
P0.4 (TX)	-	UART Transmitter
P0.5 (RX)	-	UART Receiver

Table 1: Circuit Connection between Microcontroller and Proximity Sensor

The primary wiring description include:

- **SCL (Clock) and SDA (Data)** lines connected between the EFM8BB52 and the Proximity 15 Click, with 4.7k Ohm pull-up resistors on both lines.
- **XSHUT** pin of the VL53L1X connected to a general-purpose I/O (GPIO) pin of the EFM8BB52, enabling software-controlled hardware resets of the sensor.
- **UART TX and RX** lines connected to a PC via a USB-to-Serial adapter for data visualization and debugging.
- **LED0** can be used as indicator.
- All devices were powered using regulated **3.3V** supplies from the EFM8BB52 development board.

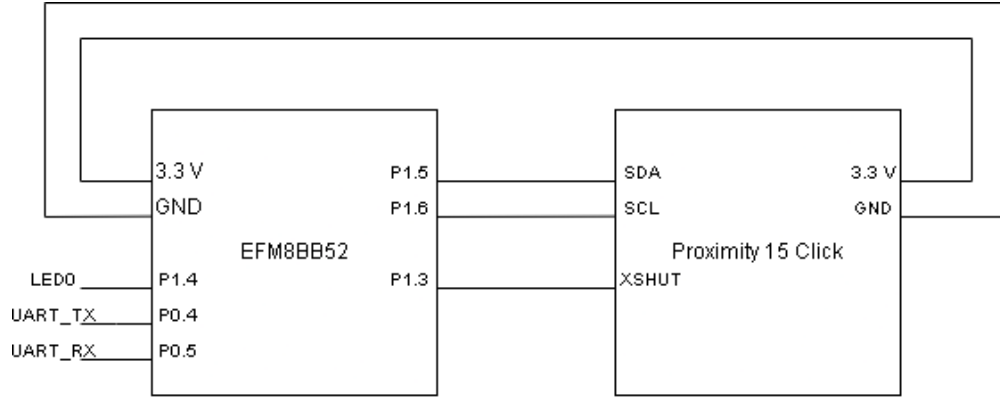


Figure 3: Circuit Connection between Microcontroller and Proximity Sensor

5.3 Software Setup

The firmware development environment and setup steps include:

- **Development IDE:** Silicon Labs Simplicity Studio 5.
- **Programming Language:** C, using Silicon Labs' Hardware Configurator for peripheral setup.
- **I2C Communication:** Custom I2C routines based on SMBus0 configuration, enabling both sensor configuration and distance data retrieval.
- **UART Communication:** UART0 configured for 115200 bps, 8-N-1 format for transmitting distance measurements to the PC terminal.
- **Testing Software:** Arduino IDE and PuTTY terminal emulators used for UART output visualization.

5.3.1 Flowchart

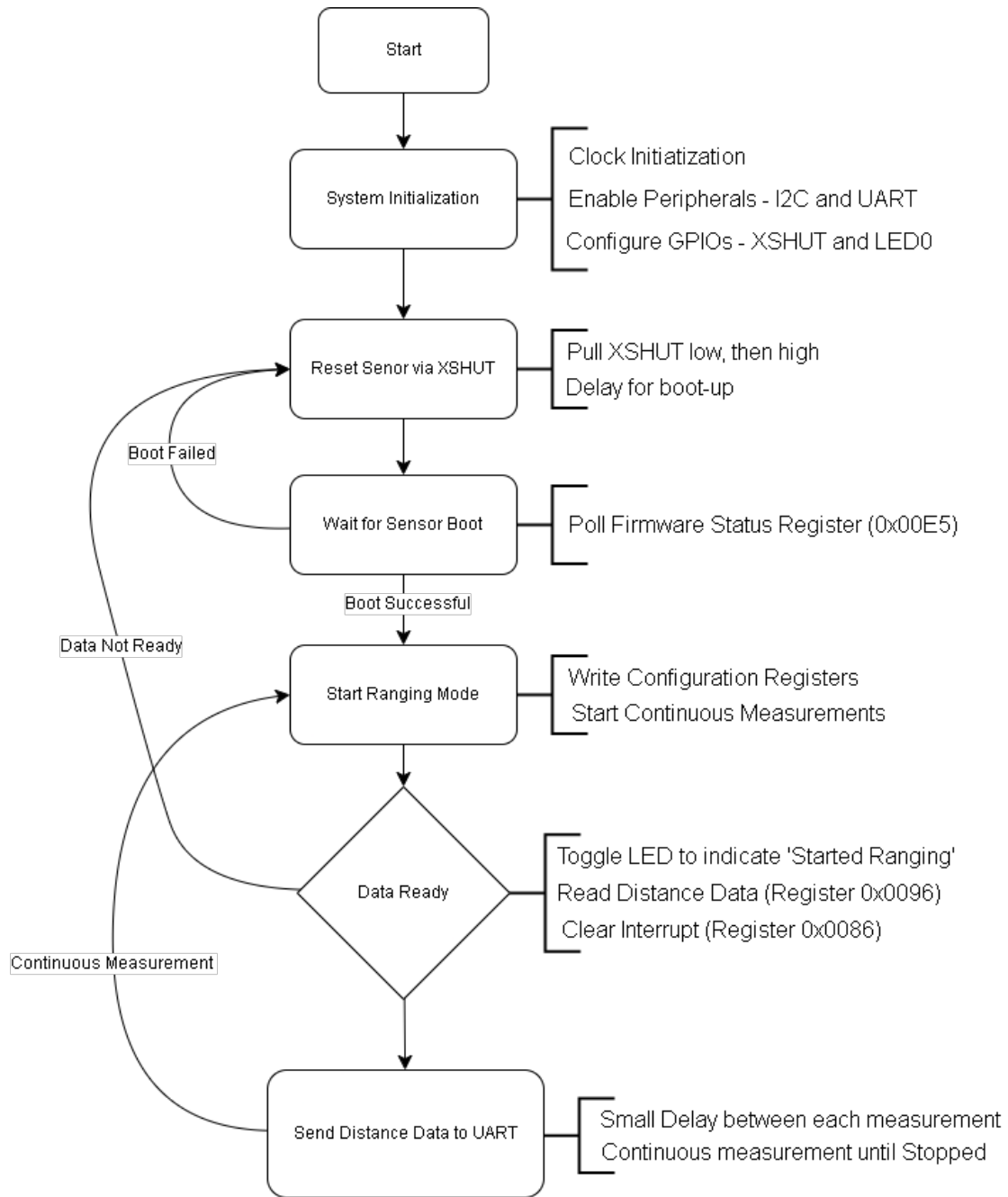


Figure 4: Flowchart Structure of the Software

6 System Design

The chapter discusses the overall design approach used to build the Software. It discusses the integration of hardware and software components, the organization of firmware modules, and key control flow algorithms necessary for the system of function reliably.

6.1 System Architecture

The system architecture is based on a modular embedded design centered around the EFM8BB52 microcontroller.

It consists of the following major blocks:

- **Sensor Layer:** The **Proximity 15 Click** board (VL53L1X sensor) continuously measures the distance to an object and communicates with the microcontroller via the I2C bus.
- **Processing Layer:** The **EFM8BB52 MCU** configures the sensor, reads distance measurements, and processes the data.
- **Communication Layer:** Processed distance data is transmitted to an external terminal (PC) over the UART serial interface for observation, logging, and analysis.
- **User Interface Layer:** A simple onboard LED toggles periodically, indicating system activity and health.

The overall system emphasizes minimal wiring, reliable communication, and real-time distance data availability.

6.2 Firmware Design

The firmware was written in C language using the Simplicity Studio 5 IDE, leveraging a modular design to separate hardware configuration, I2C communication, UART transmission, and application logic. The Firmware structure is as follows:

- **InitDevice.c/h:** Handles initialization of MCU clocks, I2C (SMBus), UART0, GPIOs (e.g., XSHUT, LED).
- **VL53L1X.c/h:** Provides required functions used to control, read and write sensor registers.
- **main.c:** Contains the main control loop, sensor boot handling, sensor configuration, data acquisition, and UART data transmission.

Special emphasis was placed on:

- Robust **error handling** (e.g., timeout detection during I2C operations).
- Efficient use of **interrupt flags** (e.g., UART transmit complete).
- Clear **delay management** to ensure hardware timing requirements (such as sensor boot delays) were met.

6.3 Key Algorithms

The essential algorithms implemented in the firmware are:

- **Sensor Boot Polling**

- After a hardware reset (XSHUT low-high toggle), the MCU repeatedly reads the VL53L1X firmware status register (0x00E5) until the boot-complete flag is set (bit 0 = 1).
- This ensures the sensor firmware is fully loaded before any further configuration.

- **Ranging Start Procedure**

- Specific registers are written to configure the VL53L1X sensor for continuous distance measurement.
- These include VHV configuration and phase calibration registers, followed by a start command.

- **Distance Data Acquisition**

- The MCU continuously polls the sensor's data ready register (0x0089).
- When new data is available, it reads the distance register (0x0096) and immediately clears the interrupt (0x0086) to prepare for the next measurement.

- **UART Data Transmission**

- Distance data is split into two bytes (MSB and LSB) and transmitted serially over UART0 to a connected PC.
- Transmission is performed without interrupts to maintain simple and robust operation in the prototype stage.

Each of these algorithms plays a vital role in ensuring the system works smoothly from sensor initialization to real-time distance reporting.

7 Experimental Setup

This chapter describes the experimental procedure used to validate the Height Measurement System. The system's performance was evaluated by comparing the measured distance values output via UART with manual measurements taken using a standard measuring tape.

7.1 Experiment with EFM8BB52 Microcontroller

Test Setup and Procedure:

- The EFM8BB52 microcontroller communicated with the Proximity 15 Click (VL53L1X) sensor using the I2C protocol.
- Terminal emulator software such as Arduino IDE or PuTTY was used to receive and display serial data.
- UART configuration:
 - Baud Rate: 115200 bps
 - Data Bits: 8
 - Parity: None
 - Stop Bits: 1
- A standard measuring tape was used to manually measure the true distance between the sensor and the target object.

Results:

- In the experimental phase, communication with the sensor via UART was successfully established.
- Distance values were printed consistently to the UART terminal.
- However, due to incomplete sensor initialization (sensor stuck at firmware boot phase), valid ranging values could not be obtained, and real distance comparison could not be fully completed.

7.2 Experiment with ESP32 Microcontroller

Test Setup and Procedure:

- The ESP32 microcontroller was connected to the sensor.
- The experiment was conducted due to failure in the sensor initialization with the EFM8BB52 microcontroller.
- Terminal emulator software such as Arduino IDE or PuTTY was used to receive and display serial data with same UART configuration as before.
- Similar measuring method was used as before.

Results:

- In the experimental phase, communication with the sensor via UART was successfully established.
- Distance values were printed consistently to the UART terminal.
- The Distance provided by the sensor were highly accurate.

The Results obtained with the ESP32 Microcontroller were highly accurate and were used for result comparison for the EFM8BB52.

8 Results and Discussion

This chapter presents the results obtained during the testing of the Height Measurement System and discusses the performance of the system based on UART outputs and manual distance validation attempts.

8.1 Results from EFM8BB52

The EFM8BB52 microcontroller served as the central control unit for the system, handling sensor communication via I2C (SMBus0) and data output over UART0.

During testing, the following outcomes were observed::

- The system successfully initialized both UART0 and SMBus (I2C0) peripherals using Silicon Labs' hardware configuration tools and custom initialization routines.
- A print-based UART output mechanism was implemented to display messages on the terminal. These messages were used to track the status of key firmware steps such as:
 - Sensor power-up and reset sequence
 - Sensor firmware boot status
 - I2C communication attempts
- The UART terminal reliably displayed printed messages, indicating that the microcontroller was executing the expected program logic. Some sensor register values were also printed to the terminal; however, these values were found to be default or incorrect and did not represent meaningful measurements.
- The VL53L1X sensor consistently failed to fully complete its internal firmware boot sequence, typically returning a constant firmware status value (e.g., 0x80) which indicates that it had not finished initializing.
- Several different approaches were tested to resolve this issue, including:
 - Manually toggling the XSHUT pin to reset the sensor hardware.
 - Experimenting with different delay values and timing sequences to allow proper firmware loading.
 - Attempting to trigger interrupts or clear configuration settings at specific boot stages.

Despite these efforts, the system was unable to obtain valid ranging data from the VL53L1X sensor. The results from EFM8BB52 testing therefore confirmed that while communication and firmware execution were stable, full sensor functionality could not be achieved within the available time.

Nonetheless, the consistent UART output and response to I2C commands demonstrated that the microcontroller's core functionality, peripheral configuration, and communication

routines were all working as intended.

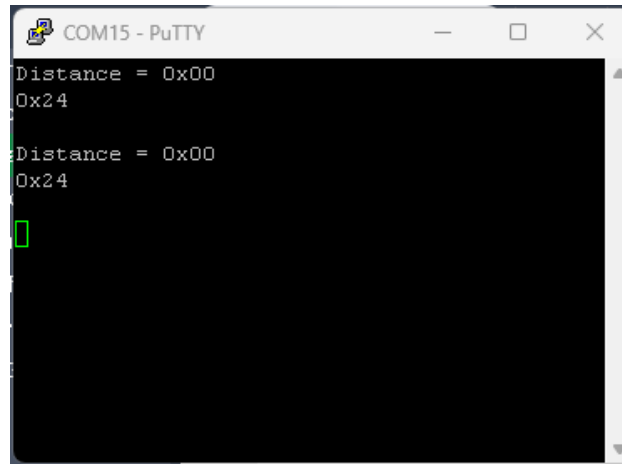


Figure 5: Terminal Output by EFM8BB52

8.2 Results from ESP32

Due to the persistent failure of the EFM8BB52 microcontroller to obtain valid distance measurements from the VL53L1X sensor, a secondary experiment was conducted using the ESP32 microcontroller.

The ESP32 was selected because of its flexible peripheral support and proven compatibility with I2C and UART-based sensor communication. The experimental setup and methodology mirrored that of the EFM8BB52 test:

- The sensor was connected to the ESP32 via the I2C interface.
- Measured distance values were printed to a serial terminal using UART.
- These UART-displayed values were then compared with physical distance measurements taken using a standard measuring tape.

Observations:

- The ESP32 successfully initialized the VL53L1X sensor and completed the firmware boot and ranging setup without issue.
- UART terminal output consistently displayed real-time distance measurements from the sensor.
- Comparison between UART-reported distance and manual tape measurements showed highly accurate and consistent results across multiple test cases.

This test validated the hardware connections, confirmed that the sensor was functional, and demonstrated that the firmware initialization process required by the VL53L1X could be achieved under the right conditions — though not with the EFM8BB52 in the current project setup. A screenshot of the UART terminal displaying real-time distance measurements is shown in Figure 6.



Figure 6: Terminal Output by ESP32

8.3 Discussion

The outcomes of both experiments highlight key differences in the performance and compatibility of the two microcontrollers when interfacing with the VL53L1X sensor.

In the case of the EFM8BB52, the system successfully established I2C communication and printed debug information via UART. However, the sensor consistently returned an incomplete firmware status (0x80), indicating that it did not fully exit the boot phase. Although a constant distance value of 36 mm (0x0024) was occasionally printed, this value did not vary with object placement and appeared to be a default or residual register value rather than an actual measured distance.

Various troubleshooting approaches — including sensor reset via XSHUT, adjusted timing delays, and manual initialization sequences — were attempted, but the sensor did not progress into an active ranging state. These results suggest that while the EFM8BB52 successfully handled communication and system control, it may lack sufficient timing precision or initialization compatibility for the VL53L1X sensor without deeper low-level driver tuning.

In contrast, the ESP32 microcontroller was able to fully initialize the sensor, start ranging operations, and output valid distance measurements. UART outputs showed dynamically updating distance values, and comparison with manual measuring tape readings confirmed a high level of accuracy. These successful results verified that:

- The sensor hardware was functional.
- The wiring and I2C bus configuration were correct.

- The issue was specific to the EFM8BB52 configuration or limitations.

This comparison demonstrates the importance of microcontroller compatibility and thorough firmware sequencing when working with complex sensors like the VL53L1X. It also validates that the system design — including UART data handling and measurement methodology — is sound and scalable, provided the initialization issue is resolved.

Overall, the experiments provided critical insight into both hardware-software integration challenges and the robustness of the sensor system under different configurations.

9 Conclusion and Future Work

9.1 Conclusion

This project focused on the development of a Height Measurement System using the Proximity 15 Click sensor based on the VL53L1X Time-of-Flight (ToF) technology, interfaced with the EFM8BB52 microcontroller.

The goal was to accurately measure distances and output the results through a UART terminal, enabling comparison with manual tape measurements.

During testing, the following milestones were achieved:

- Successful initialization and configuration of the EFM8BB52's UART and SMBus (I2C) peripherals.
- Reliable UART communication to display system status and diagnostic messages.
- Stable I2C communication between the EFM8BB52 and the Proximity 15 Click sensor at the electrical and register-access levels.

However, full functional ranging operation could not be realized with the EFM8BB52. The VL53L1X sensor remained in an incomplete firmware boot phase, returning default register values without actual distance measurements.

Subsequent experiments conducted with the ESP32 microcontroller demonstrated successful sensor initialization, accurate distance measurements, and high reliability when compared to manual measurements.

These results highlight that the hardware integration was correctly performed and that the primary limitation encountered was related to firmware sequencing and microcontroller compatibility.

Despite the challenges, the project successfully built a solid foundation for a fully functional height measurement system.

9.2 Future Work

Several important areas have been identified for future improvement and extension:

- **Enhanced Firmware Initialization:** Investigate deeper into the VL53L1X's firmware boot sequence requirements on resource-constrained microcontrollers like the EFM8BB52. Optimizing delay timings, power sequencing, and register initialization could enable successful ranging.
- **Migration to Higher-Capability Microcontroller:** Considering the success achieved with the ESP32, future implementations could shift fully to ESP32 for reliable sensor operation and to leverage additional wireless connectivity features (Wi-Fi, BLE).
- **Multi-Sensor Expansion:** Scaling the system to handle multiple VL53L1X sensors simultaneously for broader area scanning or three-dimensional height profiling.

By addressing these areas, the system can evolve into a robust, scalable, and feature-rich height measurement solution suitable for a wide range of real-world applications.

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